1.2 Exercises

September 15, 2023

1

Vertex form.

2

In intercept form, we are given p and q, the two x-intercepts of the parabola. By using the definition of the axis of symmetry, we can conclude that the x-coordinate of the vertex is $\frac{p+q}{2}$. Then, we can plug this value for x in the equation we are given, thus yielding the y-coordinate of the vertex.

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B

22

First, we find the x-coordinate of the vertex, which is the average of the two y-intercepts, that is: $\frac{6-4}{2} = 1$. Then, the x-coordinate of the vertex is $-(1-6)(1+4) = 25$. Thus, the answer is A.

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The intercept form of a parabola is $y = a(x - p)(x - q)$, which tells us that, in the given equation, the two intercepts are actually 2 and -3.

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We can solve this by multiplying the two binomials:
$y = (x + 4)(x + 3)$
$\rightarrow = x^2 + 3x + 4x + 12$
$\rightarrow = x^2 + 7x + 12$

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We can solve this like the above:
$y = (x - 5)(x + 3)$
$\rightarrow = x^2 + 3x - 5x - 15$
$\rightarrow = x^2 - 2x - 15$

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We can solve this like the above problem:
$y = 4(x + 1)(x - 6)$
$\rightarrow = 4(x^2 - 6x + x - 6)$
$\rightarrow = 4x^2 - 20x - 24$

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We can solve this by rewriting the squared binomial, and then multiplying out all of the terms:
$g(x) = -(x + 6)^2 + 10$
$\rightarrow = -(x + 6)(x + 6) + 10$
$\rightarrow = -(x^2 + 6x + 6x + 36) + 10$
$\rightarrow = -x^2 - 12x - 26$

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$y = 5(x+3)^2 - 4$
$\rightarrow = 5(x+3)(x+3) - 4$
$\rightarrow = 5(x^2 + 3x + 3x + 9) - 4$
$\rightarrow = 5x^2 + 30x + 41$

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$f(x) = 12(x-1)^2 + 4$
$\rightarrow = 12(x-1)(x-1) + 4$
$\rightarrow = 12(x^2 - x - x + 1) + 4$
$\rightarrow = 12x^2 - 24x + 16$

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Because the value of a is negative, this function has a maximum value. First, the x-coordinate of the vertex is $\frac{0+9}{2} = 4.5$, and thus, the y-coordinate of the vertex, and the maximum value, is $-12(4.5)(4.5-9) = 243$

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Because the value of a is positive, this function has a minimum value. First, the x-coordinate of the vertex is $\frac{0-15}{2} = -7.5$, and thus, the y-coordinate of the vertex, and the minimum value, is $8(-7.5)(-7.5+15) = -450$

40

Because the value of a is positive, this function has a minimum value. First, the x-coordinate of the vertex is $\frac{3+6}{2} = 4.5$, and thus, the y-coordinate of the vertex, and the minimum value, is $2(4.5-3)(4.5-6) = -4.5$

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Because the value of a is negative, this function has a maximum value. First, the x-coordinate of the vertex is $\frac{-9+4}{2} = -2.5$, and thus, the y-coordinate of the vertex, and the maximum value, is $-5(-2.5+9)(-2.5-4) = 211.25$

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Vertex form in standard form: $y = a(x-h)^2 + k$
$\rightarrow = a(x-h)(x-h) + k$
$\rightarrow = a(x^2 - xh - xh + h^2) + k$
$\rightarrow = ax^2 - 2ahx + ah^2 + k$
Hence, the x-coordinate of the vertex of this function occurs at the line $x = -\frac{-2ah}{2a} = h$, and thus, the y-coordinate of the vertex of this function is $ah^2 - 2ahh + ah^2 + k = k$, meaning the vertex of this function is (h, k).
Intercept form in standard form: $y = a(x-p)(x-q)$
$\rightarrow = a(x^2 - px - qx + pq)$
$\rightarrow = ax^2 - apx - aqx + apq$
$\rightarrow = ax^2 - (ap + aq)x + apq$
Hence, the x-coordinate of the vertex of this function is $x = -\frac{-(ap+aq)}{2a} = \frac{p+q}{2}$, and thus, the vertex of this function occurs on the line $x = \frac{p+q}{2}$

51

We are given a function with a negative value for a, and hence it has a maximum value, which is the y-coordinate of the vertex. The vertex is $(14, 6)$, and hence, the maximum height that the kangaroo can jump is 6 ft. Because we know that the vertex's x-coordinate would be halfway between where the kangaroo jumps, and where it lands, we can conclude that the kangaroo's jump is 28 ft. wide.

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The vertex of the given function is $(52.5, 45)$. Because we know that the vertex is 52.5 meters from the left end of the arch, the axis of symmetry lets us conclude that it is 52.5 meters from the right end of the arch, and thus, that the width of the arch is 105 meters.

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We are given a quadratic equation in intercept form, with the two intercepts being 0 and 160. Hence, the field's width is 160 ft. Also, the x -coordinate of the vertex is $\frac{160}{2} = 80$, and thus, the y-coordinate, and the maximum height of the field, is $-0.000234(80)(80 - 160) = 1.4976$ ft.

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The vertex of the first equation is $(6, 18)$, and thus it's maximum height is 18. The vertex of the second equation is $(6, 42)$, and hence it's maximum height is 42. Thus, the maximum height of the pogo stick with a bow spring is the larger of the two. Because these two equations are written in vertex form, we are given that the vertex is (h, k) in their respective equations. Hence, changing the constant a will not alter the vertex, and hence will not alter the maximum height. Also, because the maximum height is defined to be k, chaning h will not affect the maximum height. Thus, the only variable that affects the value of the maximum height is k.

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(a): We can tell that the x-coordinate of the vertex in this equation is $\frac{5.52+22.6}{2} = 14.06$, and that the y-coordinate of the vertex is $-0.761(14.06 - 5.52)(14.06 - 22.6) \approx 55.5$
Hence, a moisture of 14.06 contains the maximum popping volume of $55.5\frac{cm^3}{g}$
(b): We can tell that the x-coordinate of the vertex in this equation is $\frac{5.35+21.8}{2} = 13.575$, and that the y-coordiante of the vertex is $-0.652(13.575-5.35)(13.575-21.8) \approx 44.1$. Hence, a moisutre of 13.575 contains the maximum popping volume of $44.1\frac{cm^3}{g}$
(c): For the first function, the domain is $5.52 \leq x \leq 22.6$. This is because, any other x values yield negative values, which do not make sense in this situation. The range is $0 \leq y \leq 55.5$, as 55.5 is the maximum value. We can use similar reasoning for the second equation, to yield a domain of $5.35 \leq x \leq 21.8$ and a range of $0 \leq y \leq 44.1$

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We are given the vertex, as well as one point, namely $(0, 0)$. So, we can solve for a:
$0 = a(0 - 33)^2 + 5$
$\rightarrow a \approx -0.0046$
So, the equation to model this situation is $y = -0.0046(x - 33)^2 + 5$ Chaning the value of a would either make the parabola wider, or more narrow. Changing h by n units would shift the parabola to the right by n units. Changing k by m units would shift the parabola up by m units.